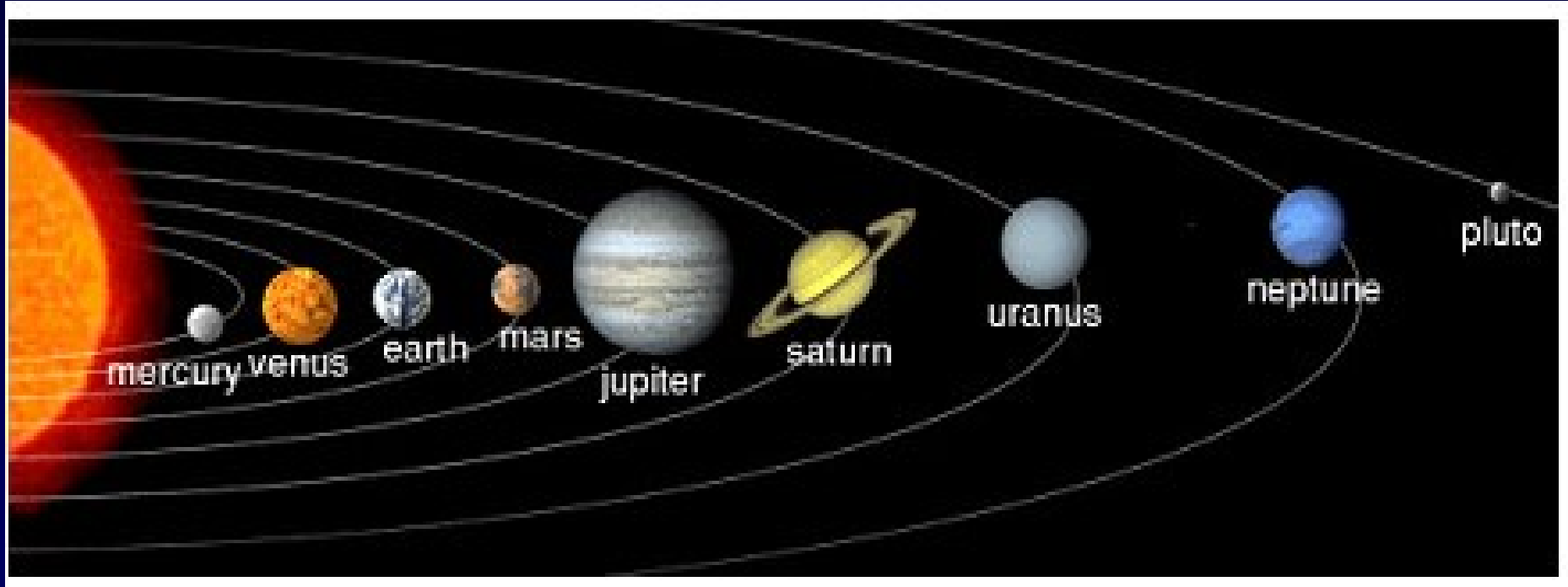


Formation of Solar System



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* Extra Classes

L08: 19 Jan. 2024

L09: 20 Jan. 2024*

L10: 20 Jan. 2024*

Journey from Geocentric to Heliocentric

- The ancient Greeks observed that several celestial bodies moved back and forth and called them the planets, meaning “wanderers”.
- In addition to the Sun and Moon, the naked eye could discern the planets Mercury, Venus, Mars, Jupiter and Saturn.
- Geometrical ideas were introduced into astronomy by the Greek philosopher Thales in the sixth century BC. This advance enabled the Greeks to develop astronomy to its highest point in the ancient world.
- Aristotle (384 - 322 BC) summarized the Greek work performed prior to his time and proposed a model of the universe with the Earth at its center.
- This geocentric model became imbedded in religious conviction and remained in authority until late into the Middle Ages. It did not go undisputed; Aristarchus of Samos (c.310 - c.230 BC) determined the sizes and distances of the Sun and Moon relative to the Earth and proposed a heliocentric (sun-centered) cosmology.

Some Definitions

Solar System

- The solar system is the gravitationally bound system of the Sun and the objects that orbit it.
- Of the bodies that orbit the Sun directly, the largest are the four gas and ice giants and the four terrestrial planets, followed by an unknown number of dwarf planets and innumerable small Solar System bodies.
- Of the bodies that orbit the Sun indirectly - the natural satellites - two are larger than Mercury and one is nearly as large.

Star: is a dense ball of gases whose surface is heated to incandescence by the energy released through nuclear fusion within the star.

Planet: is different from stars, as they do not produce their own light. They shine by reflecting light from a star. Today we know that the planets form the solar system.

Galaxy: is a huge collection of gases, dust, and billions to trillions of stars and their solar systems, all held together by force of gravity. It can have spiral, or elliptical, or irregular shape. The **Milky Way** is the galaxy that includes our Solar System.

Our Galaxy: the Milky Way Galaxy

- has about 200 billion stars, and lots of gas and dust.
- is of spiral type in shape.
- about 100,000 light-years wide.
- our Sun is halfway to the edge, revolving at half a million miles per hour around the center of the Galaxy.
- takes our Solar System about 200 million years to revolve once around our galaxy.

Meteoroids: Meteoroids are lumps of rock or iron that orbit the sun, just as planets. Meteoroids, especially the tiny particles called micrometeoroids, are extremely common throughout the solar system.

Meteorites: A meteorite is a solid piece of debris from an object, such as a comet, asteroid, or meteoroid, that originates in outer space and survives its passage through the atmosphere to reach the surface of a planet. When the original object enters the atmosphere, various factors such as friction, pressure, and chemical interactions with the atmospheric gases cause it to heat up and radiate energy. It then becomes a **meteor** and forms a fireball, also known as a **shooting star** or **falling star**. It is also known as "**bolides**" which in other words is a meteorite large enough to create an impact crater. Once it settles on the larger body's surface, the meteor becomes a **meteorite**.

Meteorites vary greatly in size. Meteorites smaller than 2 mm are classified as **micrometeorites**.

Asteroids

An asteroid is a minor planet of the inner Solar System. Historically, the term is applied to any astronomical object orbiting the Sun that did not resolve into a disc in a telescope and was not observed to have characteristics of an active comet.

Comets are cosmic snowballs of frozen gases, rock, and dust that orbit the Sun. When they pass close to the Sun, they warm and begin to release gases, a process that is called **outgassing**. This produces a visible atmosphere or **coma**, and sometimes also a **tail**. These phenomena are due to the effects of solar radiation and the solar wind acting upon the nucleus of the comet.

The comet may be up to 15 times Earth's diameter, while the tail may stretch beyond one astronomical unit. If sufficiently bright, a comet may be seen from Earth without the aid of a telescope

The main difference between asteroids and comets is their composition, as in, what they are made of. Asteroids are made up of metals and rocky material, while comets are made up of ice, dust and rocky material. Both asteroids and comets were formed early in the history of the solar system about 4.5 billion years ago.

Astronomical Unit (AU): One astronomical unit is the average distance between the Earth and the Sun. It is about 150 million km [93 million miles].

Universe

A universe contains all existing matter and space; it is also known as the cosmos. The universe is believed to be at least 10 billion light years in diameter and contains a vast number of galaxies. It has been expanding since its creation after the Big Bang about 14 billion years ago.

OUR SOLAR SYSTEM

- Our solar system consists of (i) a central star (the Sun), (ii) 8 planets orbiting the Sun, and (iii) dwarf planets, moons, asteroids, comets, meteors, interplanetary gas, dust, and all the “space” in between.
- Except for Earth, the planets of the solar system are named after Greek and Roman gods and goddesses.

Planets are categorized in two groups: inner and outer planets

- Small, rocky planets are the inner planets closest to the sun.
- Gas giants are the outer planets

- **Inner Planets:**

- Mercury
- Venus
- Earth
- Mars

- **Outer Planets**

- Jupiter
- Saturn
- Uranus
- Neptune

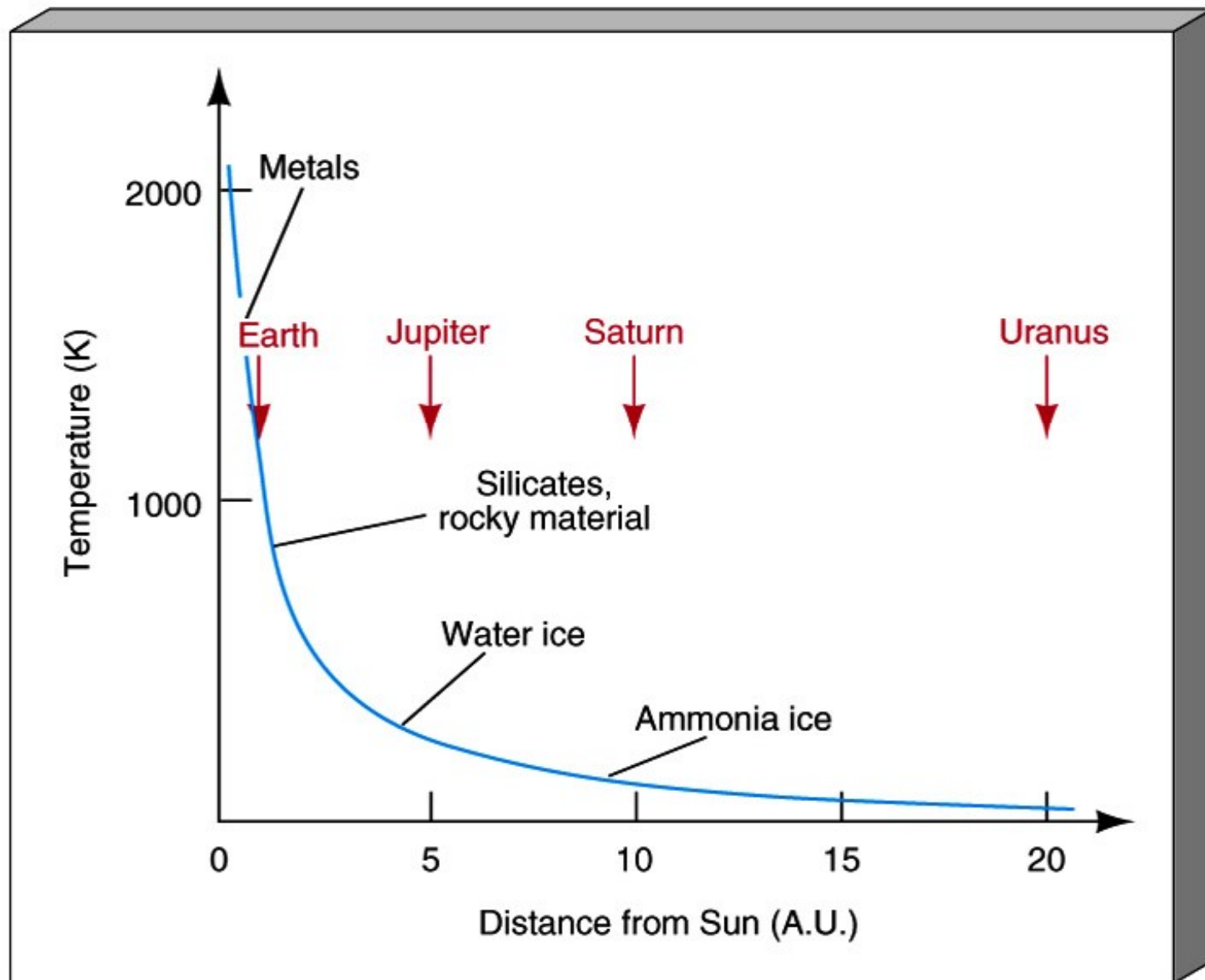
1. Characteristics of Inner Planets

- They are made up mostly of rock and metal.
- They are very heavy.
- They move slowly in space.
- They have no rings and few moons (if any).
- They have a diameter of less than 13,000 km.

2. Characteristics of Gas Giants

- They are made up mostly of gases (primarily hydrogen & helium).
- They are very light for their size.
- They move quickly in space.
- They have rings and many moons.
- They have a diameter of 49,000 km or greater.

Temperature of Planets of Solar System



What Keeps Our Planets & Other Objects In Space In Orbit??

GRAVITY & INERTIA

- **Gravity** – A force that pulls all objects toward each other.
- **Inertia** – The tendency of an object to stay either at rest or in motion along a straight path.

The Sun

- The Sun is a bright star made of extremely hot gasses.
- The Sun's energy comes from nuclear fusion (a process in which two light nuclei merge to form a single heavier nucleus) within its core and release of lots of energy in the form of heat and light.
- It is worth to remember that Stars produce light and planets reflect light.
- A Star's temperature determines its color. The coldest stars are red and hottest stars are blue.
- The Sun plays a unique role in supporting life on the planet Earth.

The Origin of the Universe and Solar System

- Big Bang formed universe over 15 Billion Years Ago (Bya). **Big Bang Theory** explains the origin of the universe. The theory is most widely accepted among scientists at the present.
- Earth formed from gases & dust ~4.6 – 5.0 Bya. The **Nebular Hypothesis** is an explanation for the formation of solar system.



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The Big Bang Theory

- The Big Bang Theory states that, in the beginning, the universe was all in one place.
- All of its matter and energy were squished into an infinitely small point, a singularity (infinitely small size, infinite density, infinitely high temperature).
- Nothing existed before this singularity → No time, No space, No Matter.
- Then it exploded.
- The tremendous amount of material blown out by the explosion eventually formed the stars and galaxies.
- Out of nothingness, came everything in the universe and it started to expand.
- After about 10 billion years, our solar system began to form.

Origin of the Universe

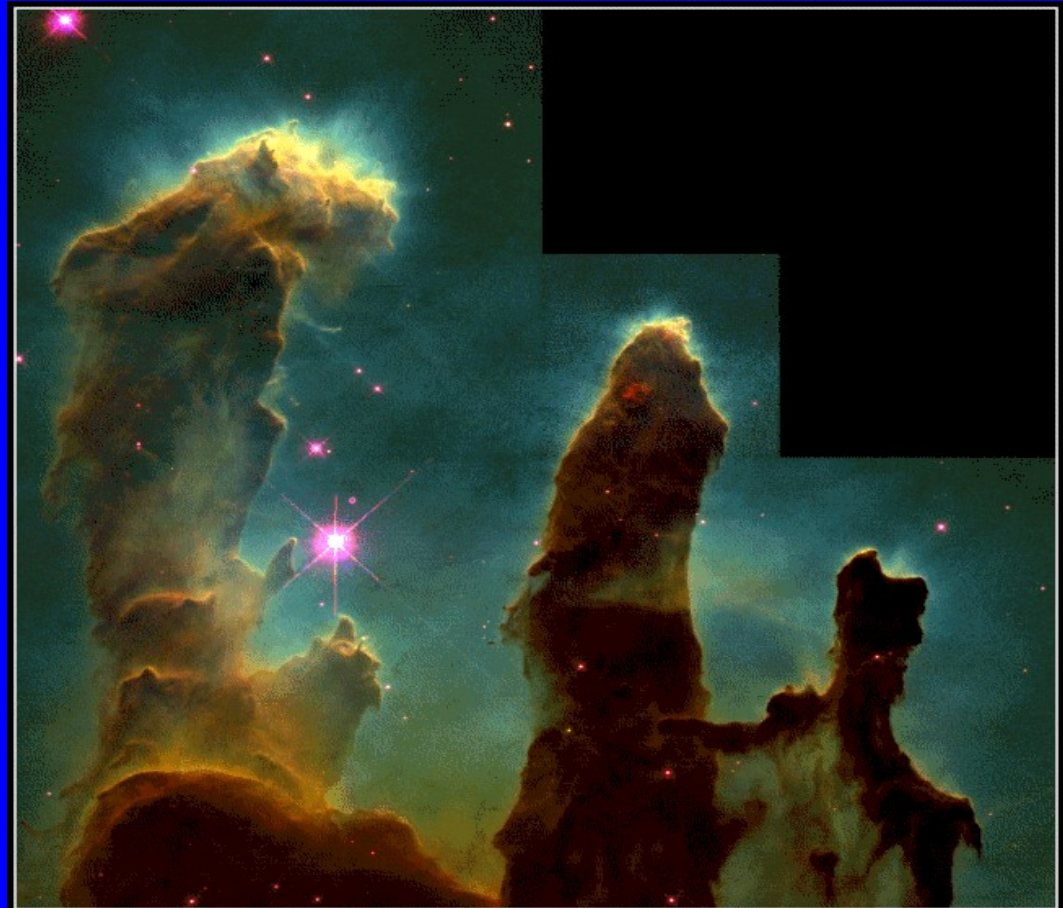
Scientists believe that the universe began in a tremendous release of energy known as the Big Bang. The universe has been expanding ever since. Evidence for this comes from:

- Observation of galaxies flying away from us in all directions.
- Remnant of the Big Bang known as the Cosmic Microwave Background Radiation.
- Observed abundance of light elements made in the Big Bang matches predictions.

Expansion implies that the universe was smaller in past.

Star Formation

Eventually the universe cooled and atoms formed. With the help of gravity these atoms (mostly hydrogen) clumped together to form stars.



Gaseous Pillars · M16

HST · WFPC2

PRC95-44a · ST ScI OPO · November 2, 1995
J. Hester and P. Scowen (AZ State Univ.), NASA

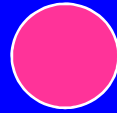
ATOMS

PROTON



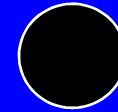
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NEUTRON



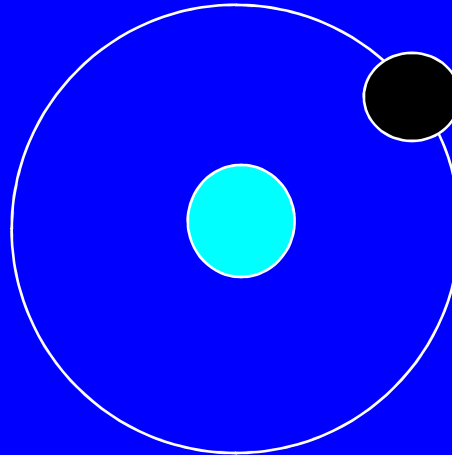
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ELECTRON

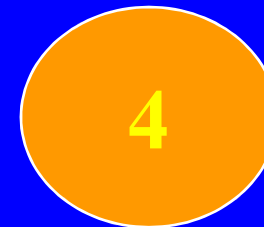
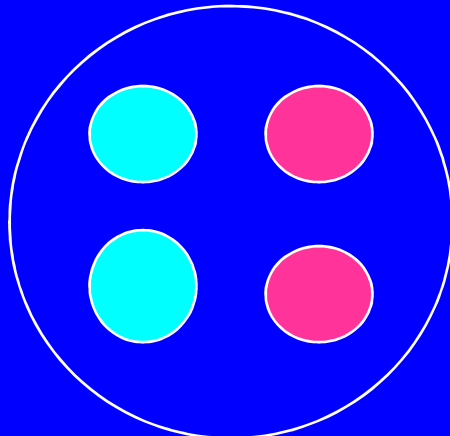


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HYDROGEN ATOM (H)

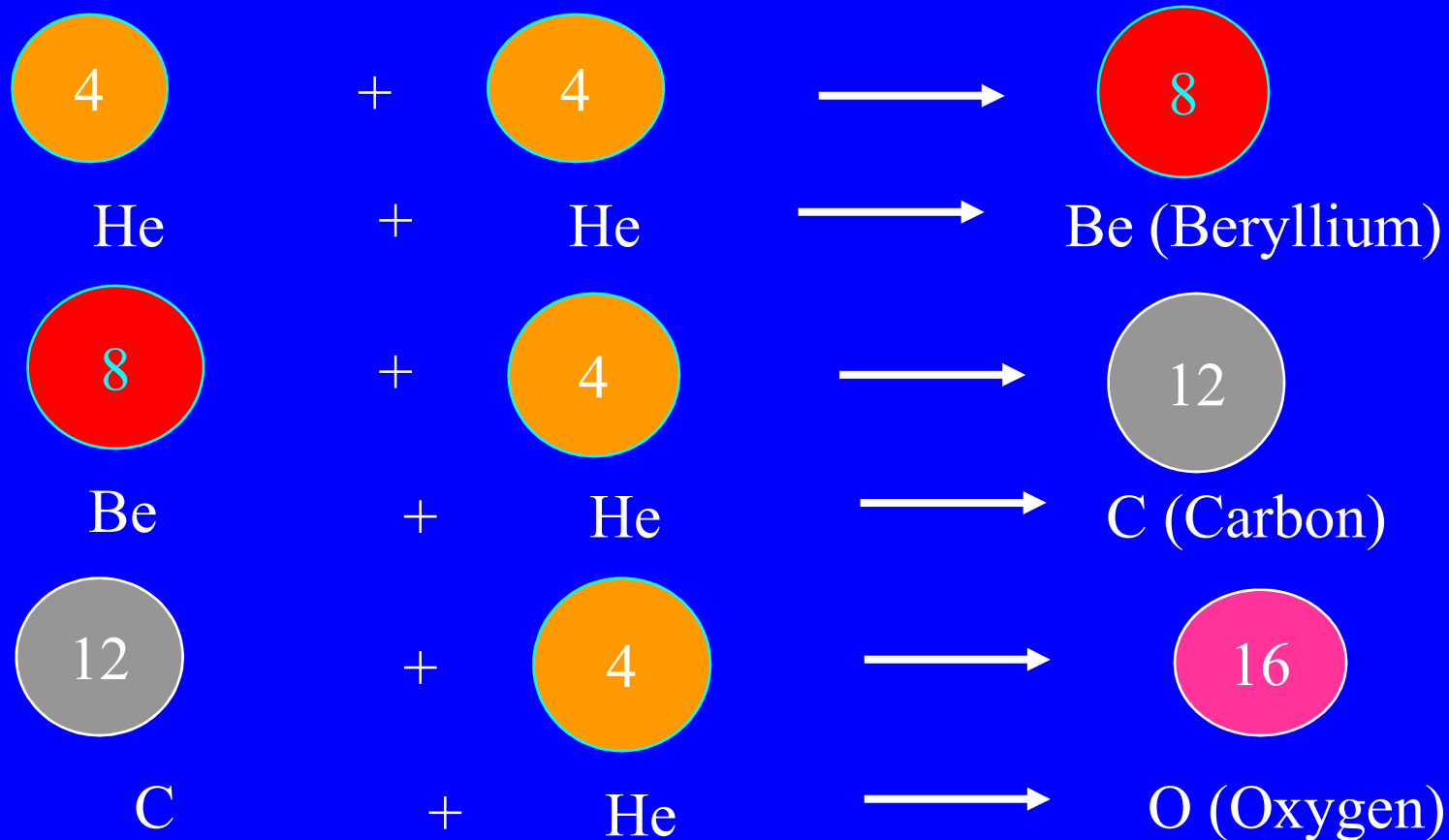


Combine 4 hydrogen atoms to make a helium atom. This is called fusion. It powers the sun.



HELIUM ATOM (He)

NUCLEOSYNTHESIS: Making heavier atoms by combining lighter atoms. This occurs in stars and is called fusion. This is where stars get the energy to shine.



Elements are made in stars.

Distance between Stars

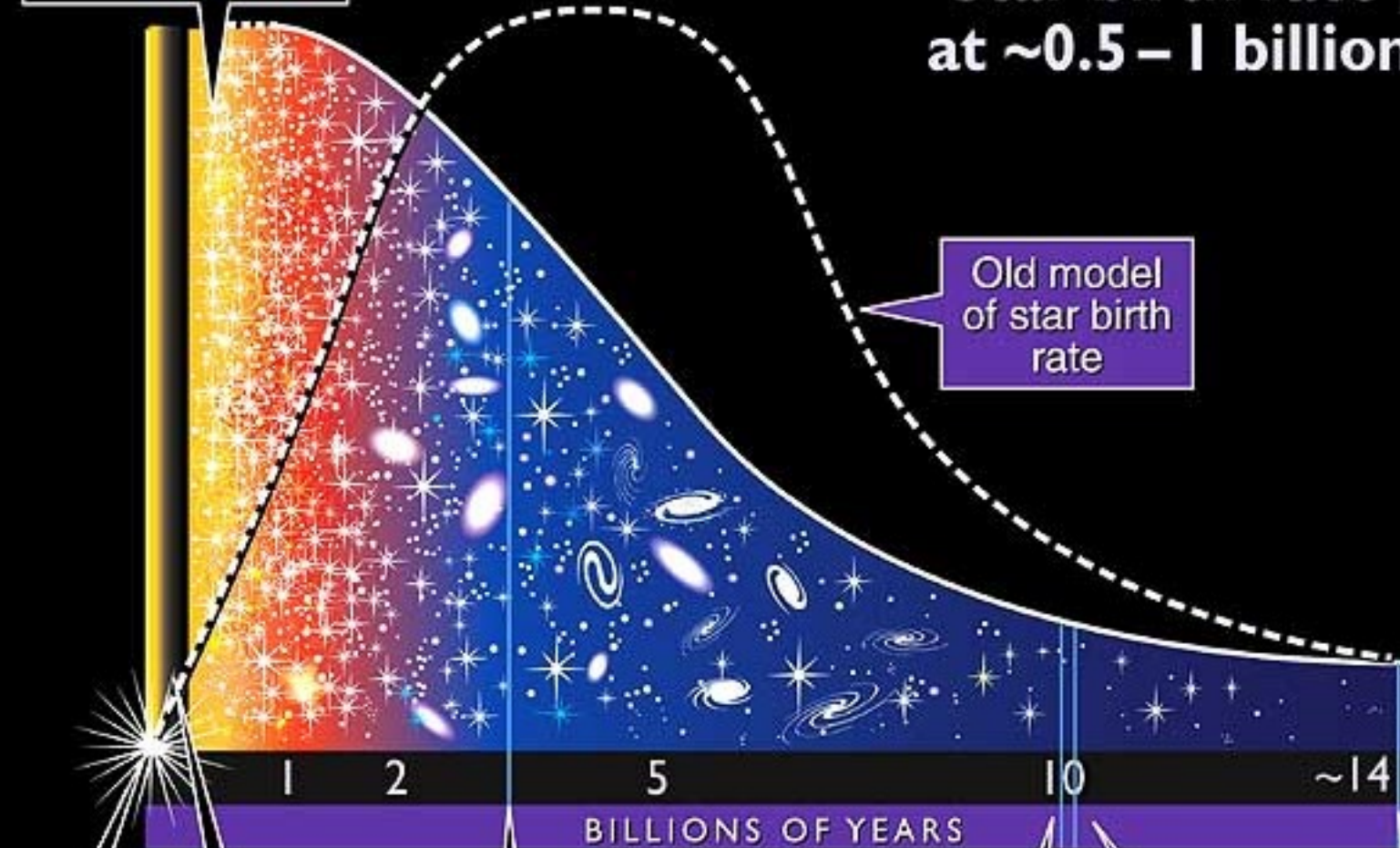
Fortunately for us, the average distance between stars is about 3 light years or 20 trillion miles. If stars were 10 times closer (~2 trillion miles), a nearby star could pull the earth into an eccentric orbit and kill life here. If the stars were much farther apart, then there would not be enough heavy elements to make life. Heavy elements (like carbon and oxygen) are the ashes of dead stars and living things need these elements.



Star birth
begins

New model:
Star birth rate peaks
at ~0.5 – 1 billion years

Old model
of star birth
rate



Big
bang

Dark
era

Milky Way
galaxy forms

Our Solar
System
forms

Life
appears
on Earth

Humankind
evolves

The Origin of the Solar System

- It is believed from studies of meteorites – remnants of the early Solar System – that our Sun and planets formed about 4.6 billion years ago from what is called a giant molecular cloud. About 4.6 billion years ago, a giant cloud floated in one of the spiral arms of the Milky Way galaxy. This cloud, called a nebula by astronomers, was made up of dust and gas, mostly hydrogen and helium, with a small percentage of heavier atoms.
- The nebular hypothesis is the most widely accepted model in the field of cosmogony to explain the formation and evolution of the Solar System. The theory was developed by German philosopher Immanuel Kant and published in his Universal Natural History and Theory of the Heavens (1755) and then modified in 1796 by Pierre Laplace - a French Mathematician.
- Originally applied to the Solar System, the process of planetary system formation is now thought to be at work throughout the universe.

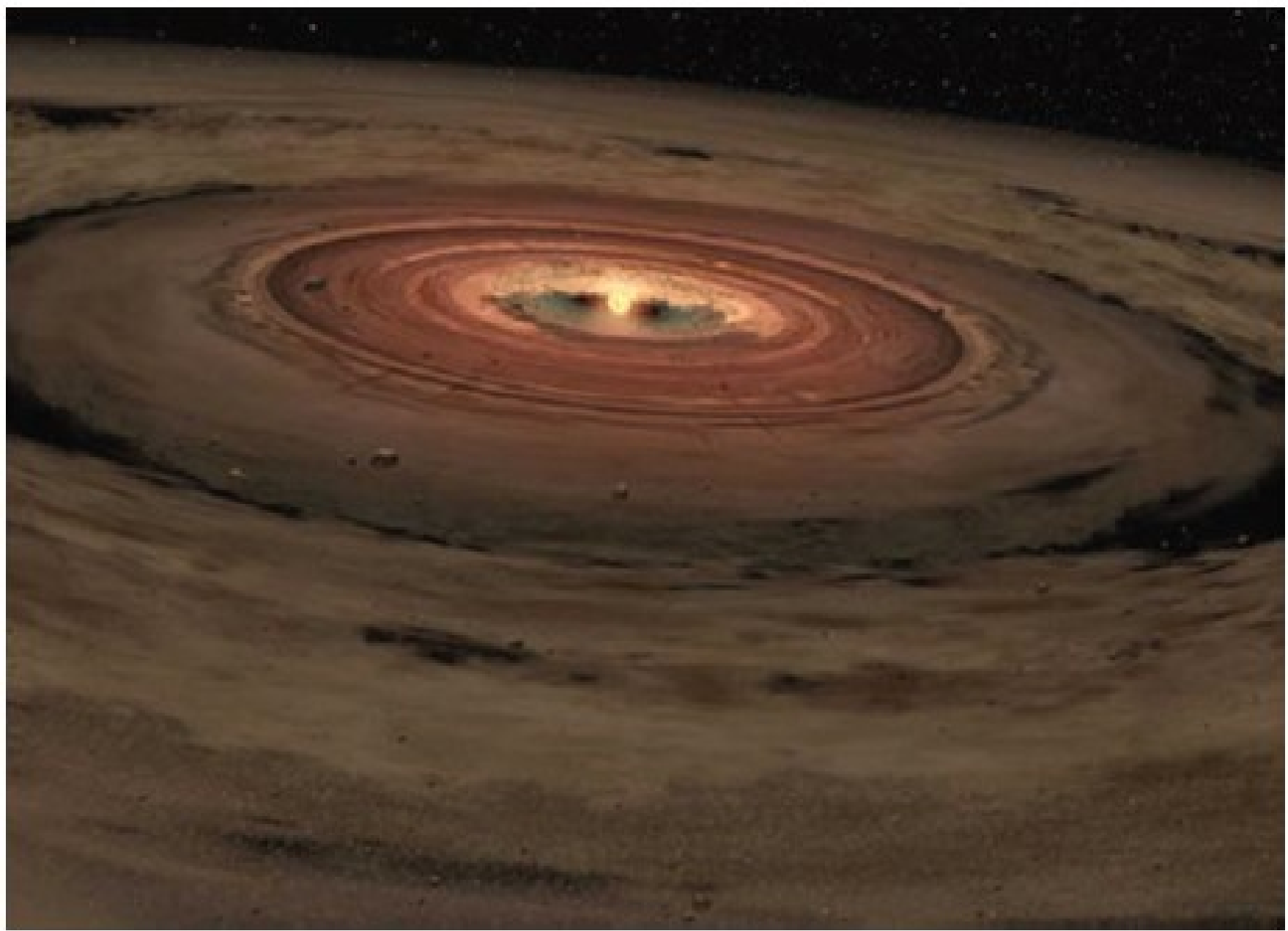
The Nebular Hypothesis

- All bodies in the solar system formed from nebula which was an interstellar cloud of gases & space dust consisting mainly Hydrogen and Helium. The interstellar gas & space dust were scattered at varying densities between the stars in our galaxy.
- This large gas cloud (nebula) began to condense. Most of the mass was in the center, there was turbulence in the outer part.
- Gravitational attraction causes the mass of gas and dust to slowly contract and it began to rotate. It is likely that the solar nebula would have had some rotational energy. As the nebula collapsed, conservation of angular momentum meant that it would have begun to spin faster and, as it became denser. Collisions within it would cause the gas to heat up.
- The dust and matter slowly falls towards the center.

In the interstellar medium, dust and gas coalesces into clouds

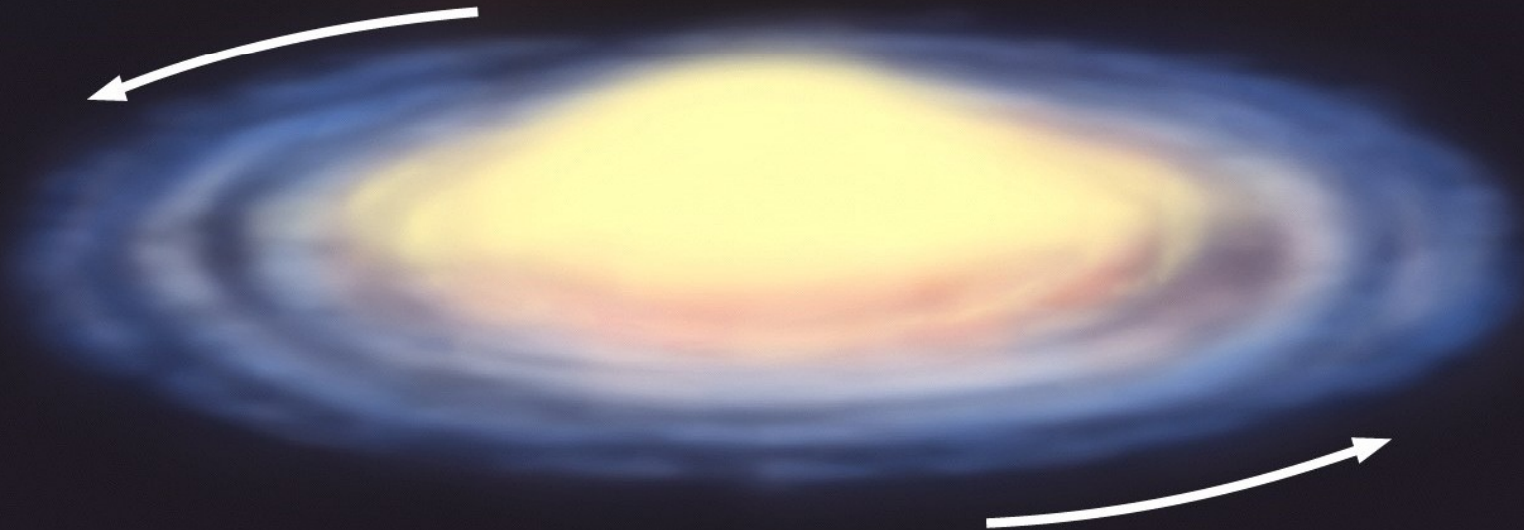


Solar Nebula



Formation of Pro-Sun

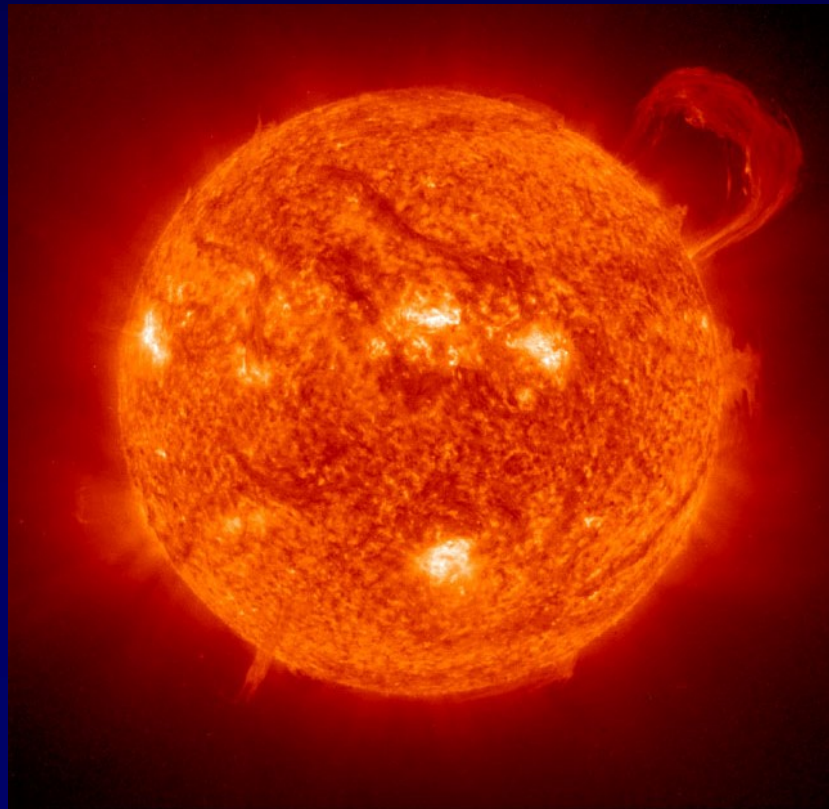
As a result of contraction and rotation, a flat, rapidly rotating disk forms with the matter concentrated at the center that became the proto-Sun.



The Sun

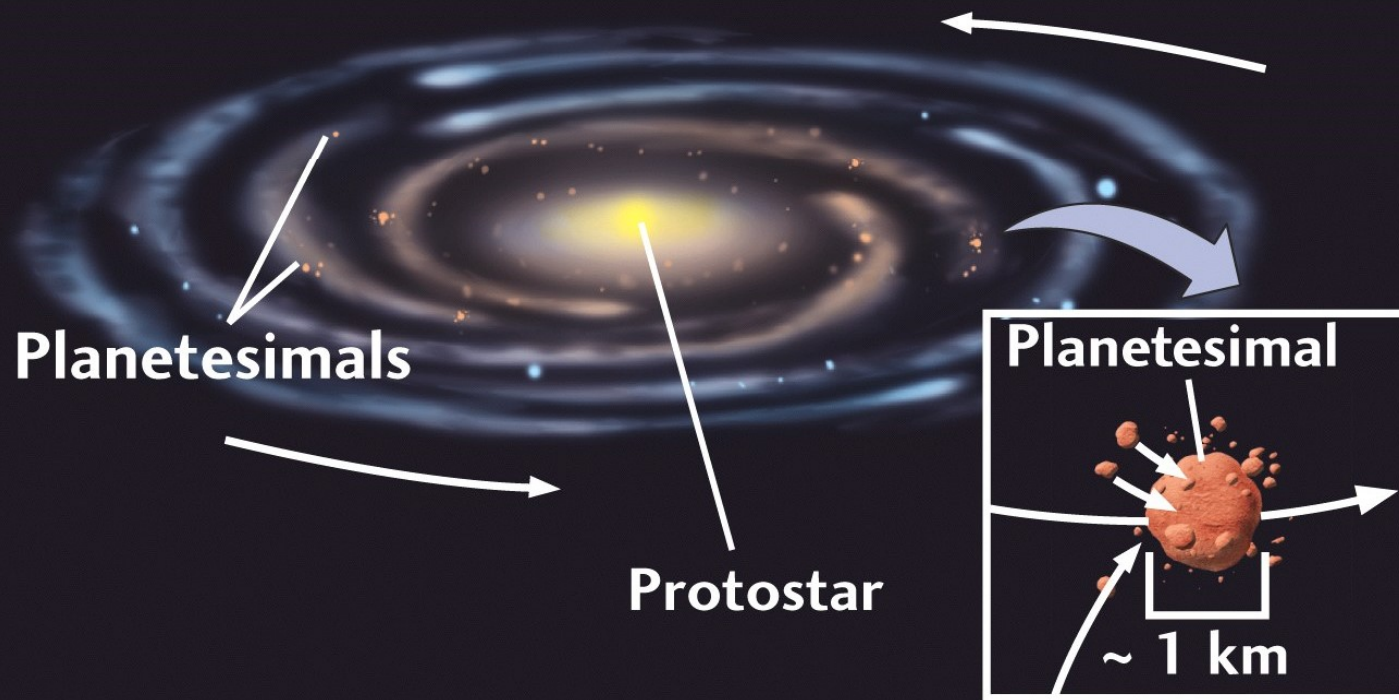
- After sufficient mass and density was achieved in the Sun, the temperature rose to one million °C, resulting in thermonuclear fusion.

H atom + H atom = He atom + energy



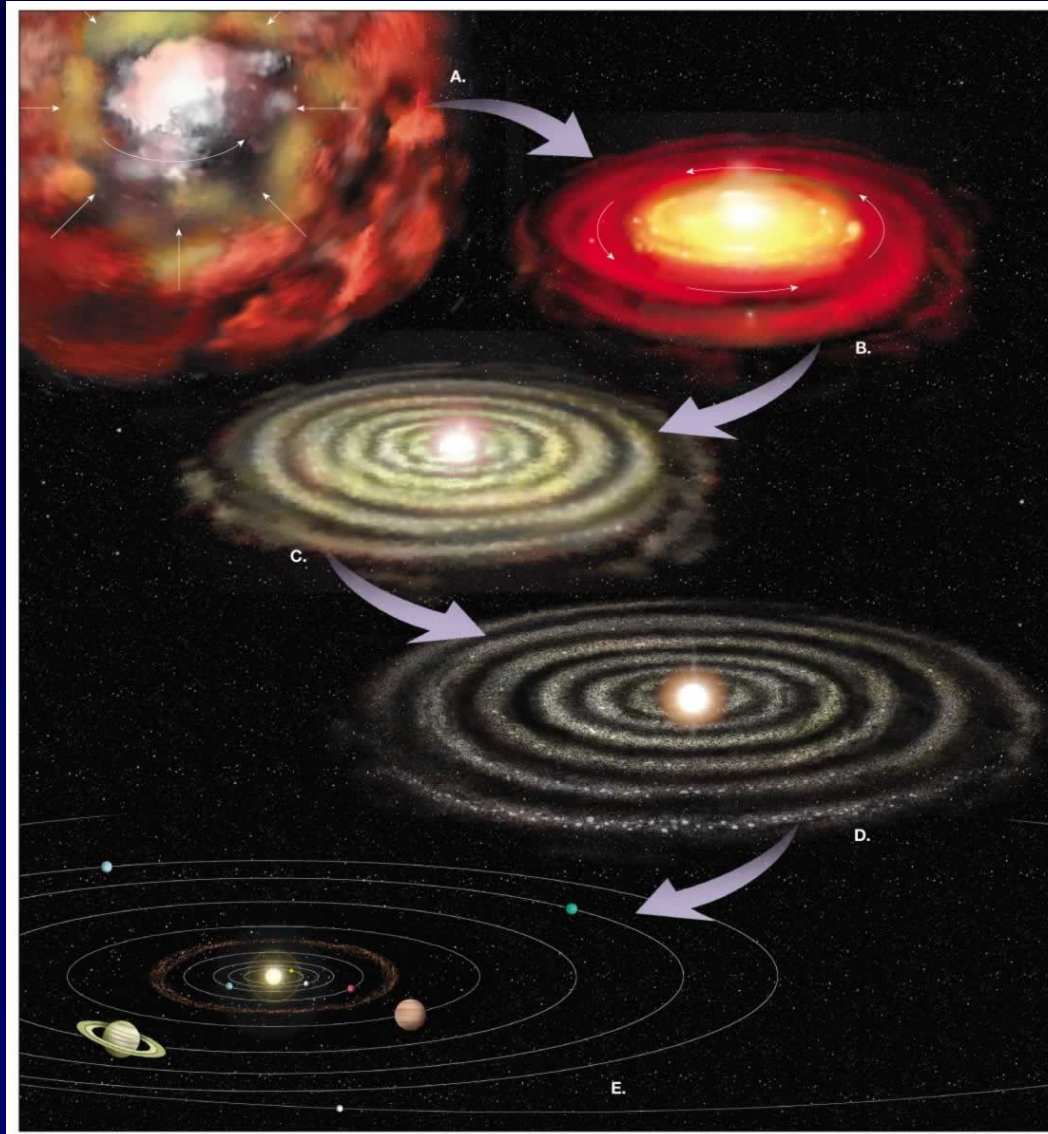
Birth of the Solar System

The enveloping disk of gas and dust forms grains that collide and clump together into small chunks or planetesimals.



Terrestrial planets form by accretion of solids

Dust → rocks → planetesimals → embryos → planets



Protoplanets

Gravitational forces allow the inner planets to accrue and compact solid matter (including light and heavy atoms)

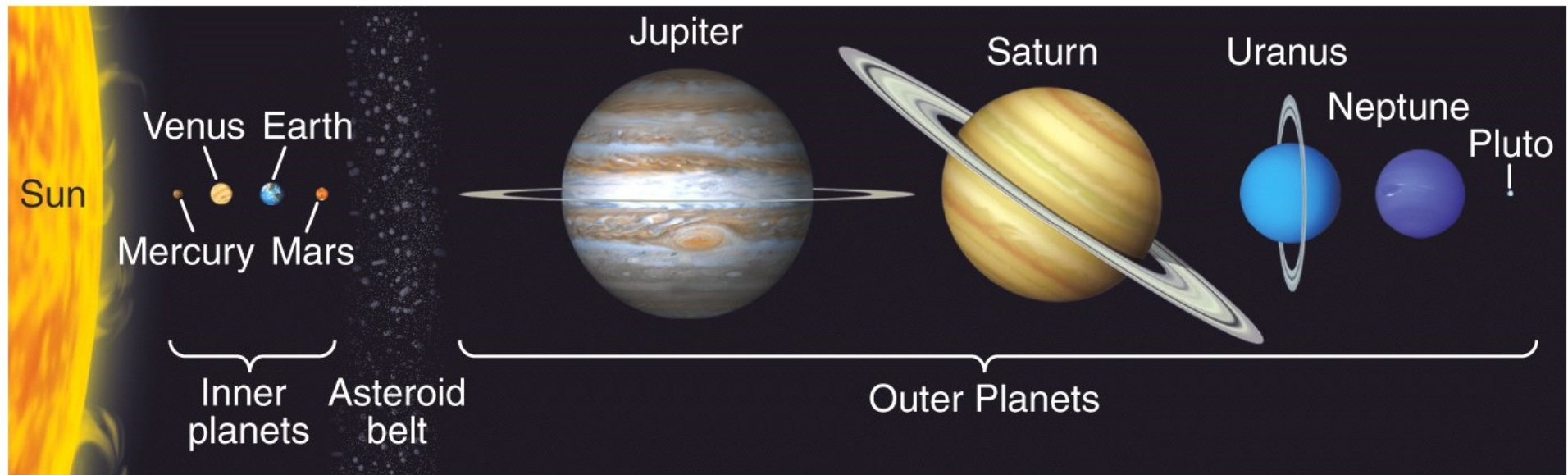
Solar radiation blew gases (primarily hydrogen, helium) away from inner planets

These gases were collected and condensed into the gas giants (Jupiter, Saturn, Uranus, Neptune)

Beyond Neptune, ice and frozen gases form Pluto, Sedna and the Kuiper Belt Objects

Left-over debris form comets and asteroids

Size of the Planets in our Solar System



Other Residents of the Solar System

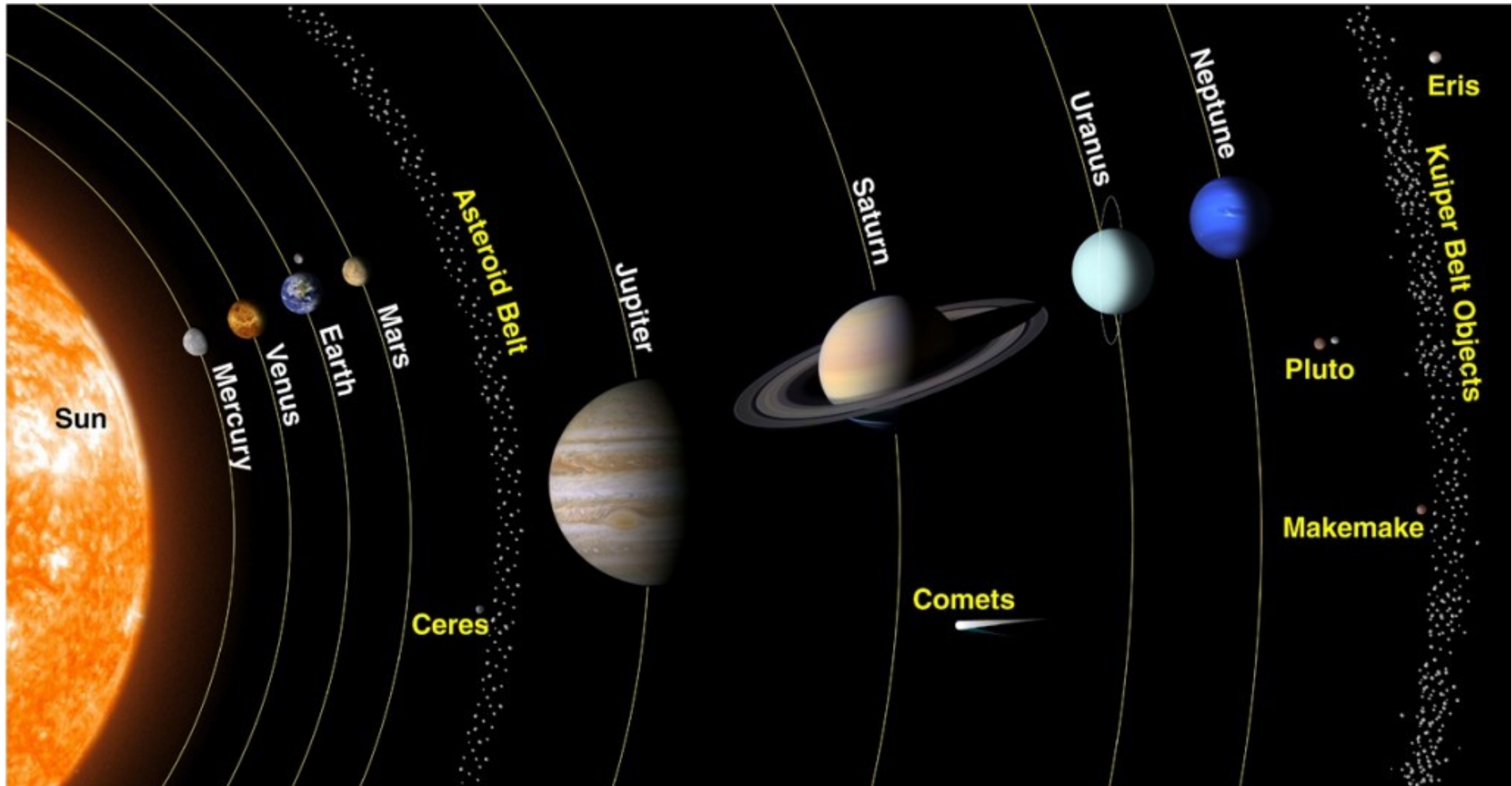
1. Dwarf planets

- diameter = 1000-3000 km,
- smaller than Moon,
- orbit the sun

The authority on the naming and classification of celestial objects, the International Astronomical Union, officially recognizes five dwarf planets in the solar system:

- 1. Pluto.**
- 2. Eris.**
- 3. Ceres.**
- 4. Makemake.**
- 5. Haumea.**

Solar System Containing Dwarf Planets



Other Residents of the Solar System

2. Asteroids

- rocky,
- $d < 1000$ km,
- orbit the sun



Other Residents of the Solar System

3. Comets

- rock & ice,
- wide range of sizes (~10 m to 100 km)
- Orbit the sun



C/2006 P1 McNaught 2007 01 20
3 X 30 sec, 85mm f/1.6
Copyright Gordon Garradd

Other Residents of the Solar System

4. Moons

- orbit planets,
- some are larger than Mercury

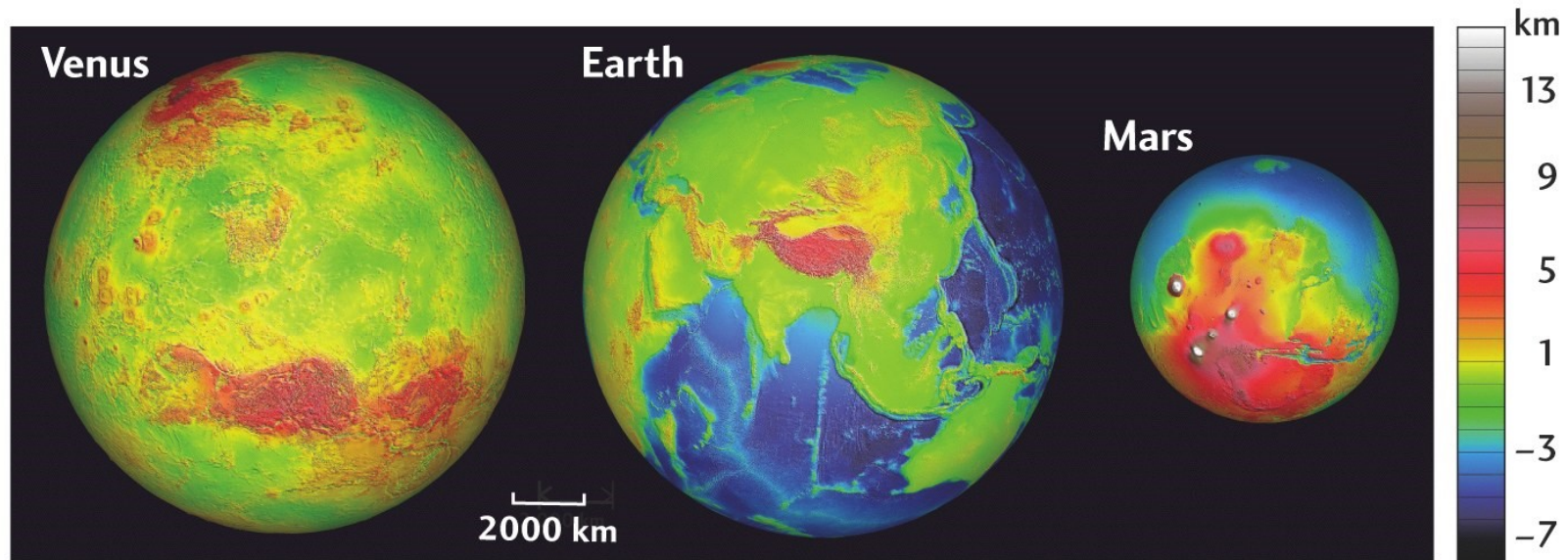
Other Residents of the Solar System

5. Meteoroids

- small fragments of asteroids that enter earth's atmosphere (dust to boulder sized)



Venus, Earth and Mars



These maps are color coded to display different elevations on the surface of each planet

Sun-Planet Distance (Relative to Earth: AU)

Mercury	0.4 AU
Venus	0.7
Earth	1.0
Mars	1.5
Jupiter	5.2
Saturn	9.5
Uranus	19
Neptune	30

1 AU = 150 million km

Overall properties of the Sun

Diameter

The Sun subtends an angle of ~ 30 arcmin in the sky. Given its distance from us of 150 000 000 km one can directly estimate the Sun's diameter:

$$D = R\theta$$

Where, θ is in radians and R is the Earth-Sun distance.

$$\theta = 30/(60 \times 57.3) \text{ rad}$$

$$= 8.7 \times 10^{-3} \text{ rad}$$

$$D = 1.5 \times 10^8 \times 8.7 \times 10^{-3} \text{ km}$$

$$= 1\,308\,900 \text{ km}$$

This is reasonably close to the accurate value of 1 391 978 km. The Sun's radius is thus $\sim 700\,000$ km.

Mass

The mass of the Sun can be derived using the same methods that Newton employed with the Moon. Equating the force attracting the Earth to the Sun given by Newton's Law of Gravitation with that due to centripetal acceleration we get:

$$MmG/R^2 = mv^2/R \rightarrow M = v^2R/G$$

where M is the mass of the Sun, m is the mass of the Earth, v is the velocity of the Earth around the Sun, G is the universal constant of gravitation and R is the distance of the Earth from the Sun.

Substituting $v = 2\pi R/P$ (where P is the period of the Earth's orbit) in above equation, we get:

$$M = 4\pi^2 R^3 / GP^2$$

$$\begin{aligned} &= 4 \times (3.14159)^2 \times (1.496 \times 10^{11})^3 / (6.67 \times 10^{-11} \times (3.156 \times 10^7)^2) \text{ kg} \\ &= 1.99 \times 10^{30} \text{ kg} \end{aligned}$$

Density

From the radius and mass of the Sun, one can easily derive its average density:

$$\begin{aligned}\text{Volume} &= (4/3)\pi r^3 \\ &= (4/3)\pi(700\,000\,000)^3 \\ &= 1.4 \times 10^{27} \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Therefore Density} &= M/V \\ &= (2 \times 10^{30}) / (1.4 \times 10^{27} \text{ kg m}^{-3}) \\ &= 1428 \text{ kg m}^{-3}\end{aligned}$$

This is about 40% greater than that of water and about 26% that of the Earth.

The Sun's Total Energy Output

Let us assume one has a value, e , for the Sun's energy falling on 1 m^2 of the Earth's atmosphere. This is just 1 m^2 of a spherical surface centred on the Sun which has an area of:

$$A = 4\pi(1.5 \times 10^{11})^2 \text{ m}^2$$

where $1.5 \times 10^{11} \text{ m}$ is the distance of the Earth from the Sun. As all of the Sun's energy must pass through this surface, the Sun's total energy output is simply $A \times e$. Since the measured value of e is 1370 W falling on 1 m^2 area, the Sun's total energy output is thus:

$$\begin{aligned} E &= 1370 \times 4\pi(1.5 \times 10^{11})^2 \text{ W} \\ &= 3.86 \times 10^{26} \text{ W} \end{aligned}$$

Protoearth

- Larger than Earth today
- Homogeneous composition
- Bombarded by meteorites
- Moon formed from collision with large asteroid
- Pieces incorporated into the protoearth

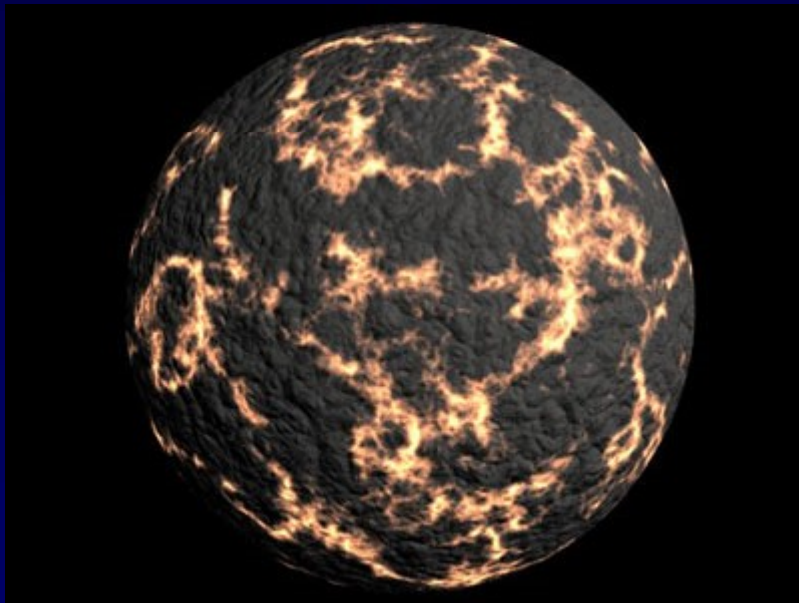


Protoearth.....Contd.

- Radioactive heat
- Spontaneous disintegration of atoms
- Fusion reactions → heavier elements
- Heat from contraction (protoplanet shrinks due to gravity)
- Protoearth partially melts
- Denser materials move to center
- density stratification (layered Earth)
- Highest density material at center (core)
- Lowest density material at surface (crust)
- Chemical composition and physical properties of layered earth

The Early Earth Heats Up

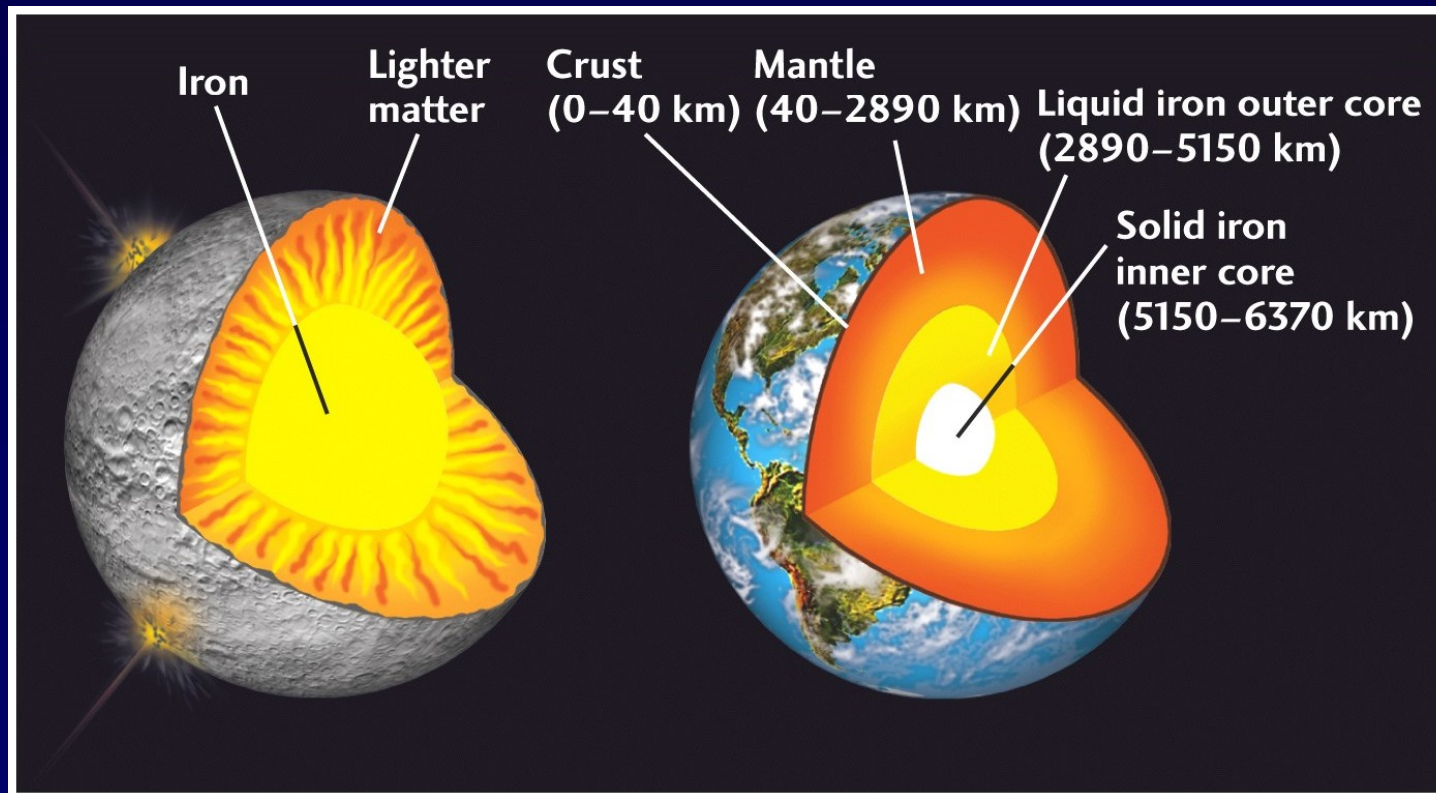
3 major factors that caused heating and melting in the early Earth's interior:



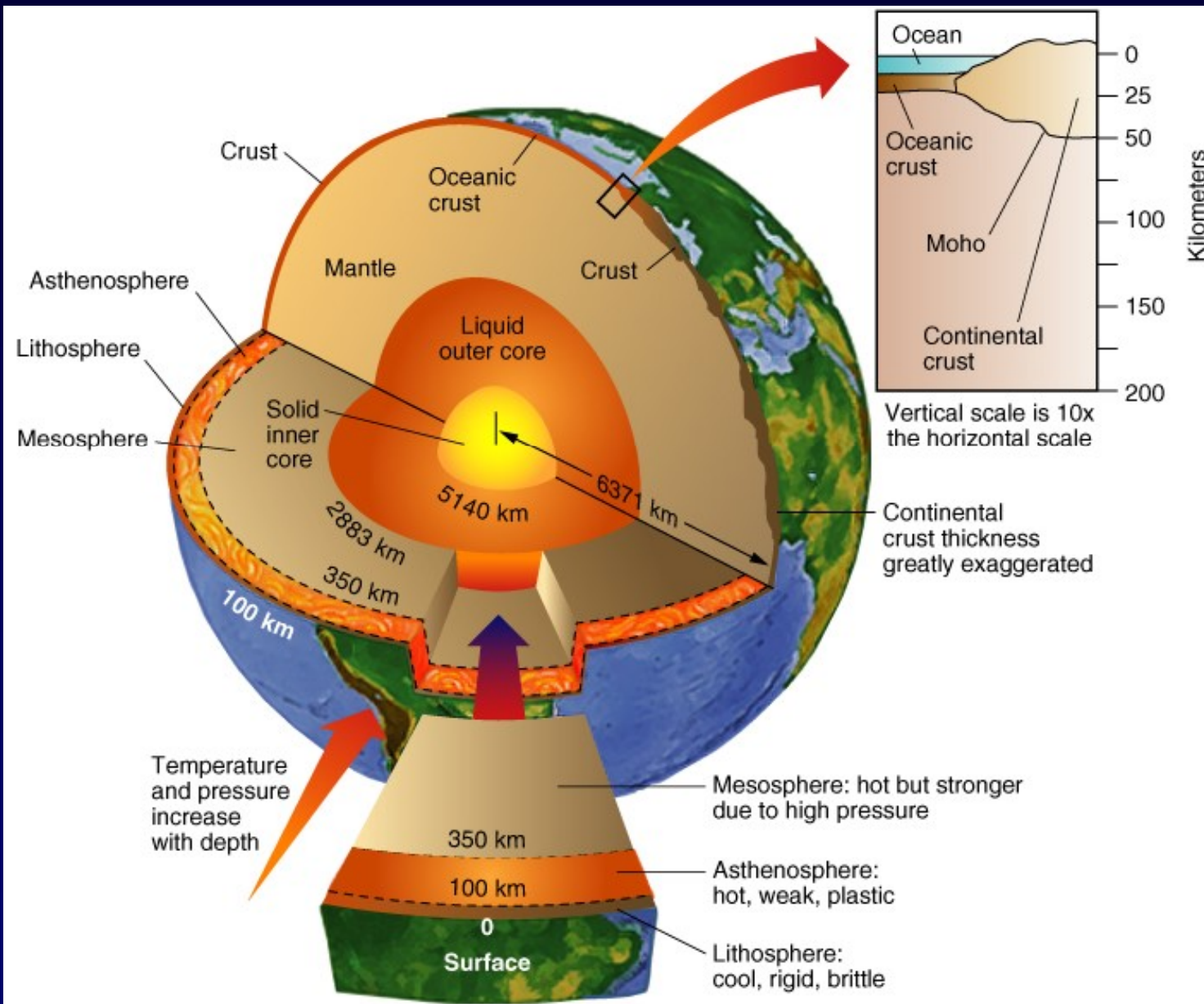
1. **Collisions (Transfer of kinetic energy into heat)**
2. **Compression**
3. **Radioactivity of elements (e.g. uranium, potassium, or thorium)**

Global Chemical Differentiation

- In a process called global chemical differential, the heavier elements, including the melted iron, began to sink down into the core of the Earth, while the lighter elements such as oxygen and silica floated up towards the surface.
- Chemical separation was completed by about 4.3 billion years ago, and the Earth had developed a inner and outer core, a mantle and crust



Gross Structure of the Earth



Oceanic Crust

Basaltic
SiMa
Denser
Thinner
Mostly below sea
Surface
Much younger in age

Continental Crust

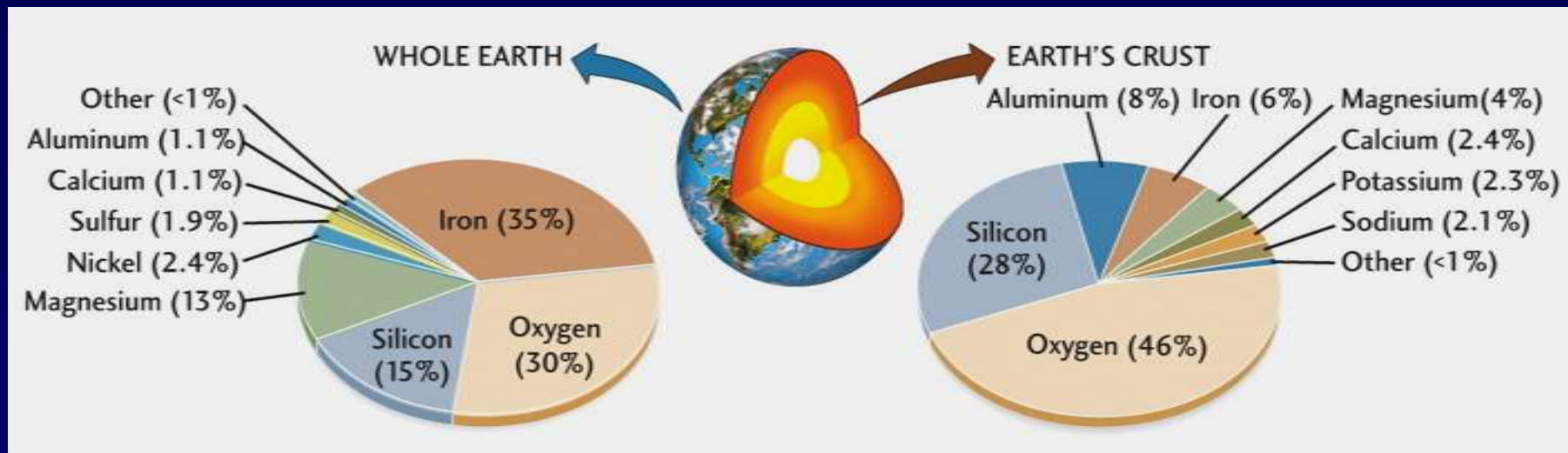
Granitic
SiAl
Lighter
Thicker
Mostly above sea
Surface
Much older in age

Chemical Composition of Earth

Each of the major layers has a distinctive chemical composition, with the crust being quite different from the Earth as a whole

Whole Earth:
 $\text{Fe} + \text{O} + \text{Si} + \text{Mg} = 93\%$

Crust:
 $\text{Si} + \text{O} + \text{Al} = 82\%$



Formation of the Moon

The Giant Impact Hypothesis –

~ 50 million years after the initial creation of Earth, a planet about the size of Mars collided with Earth

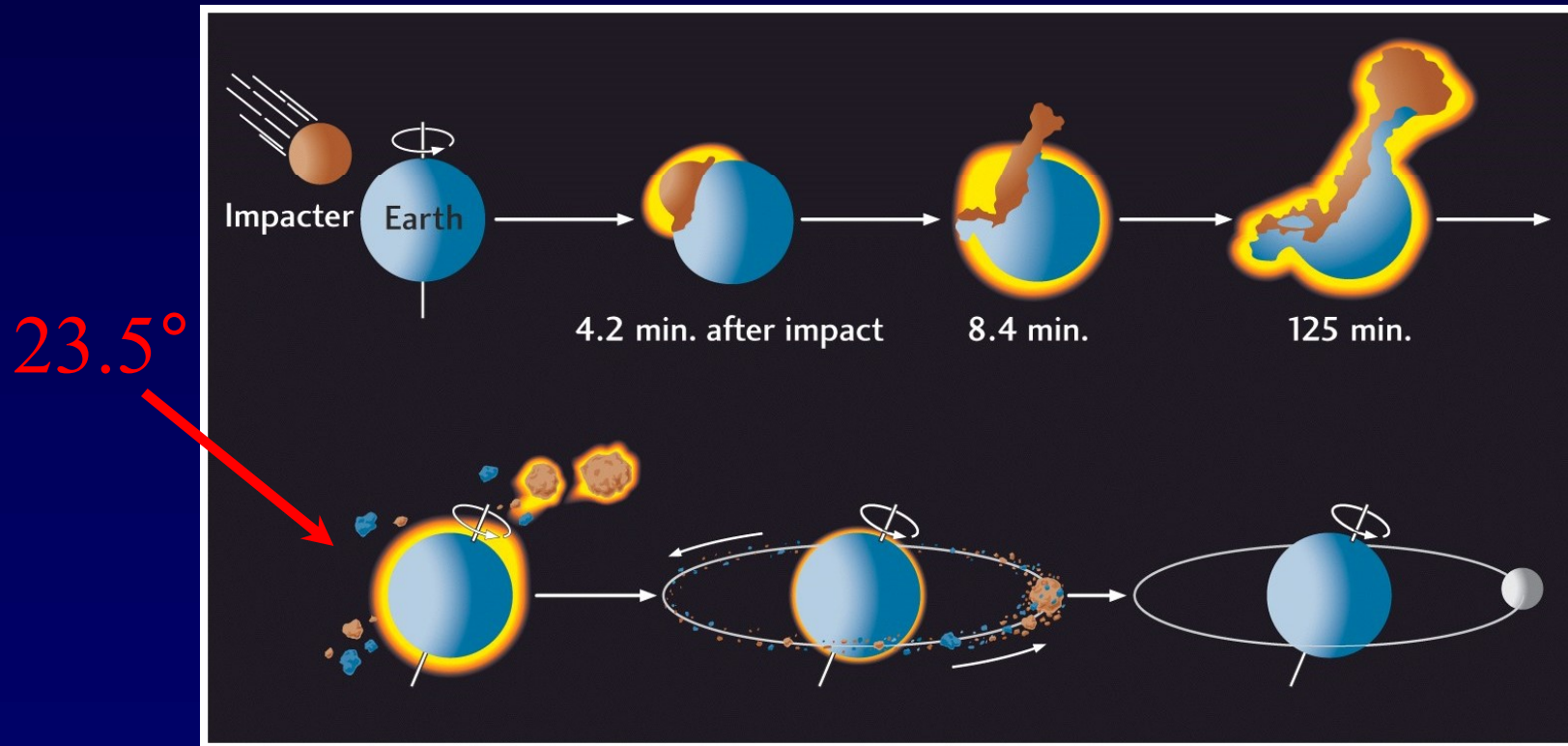
This idea was first proposed about 30 years ago, but it took calculations by modern high-speed computers to prove the feasibility



Formation of the Moon

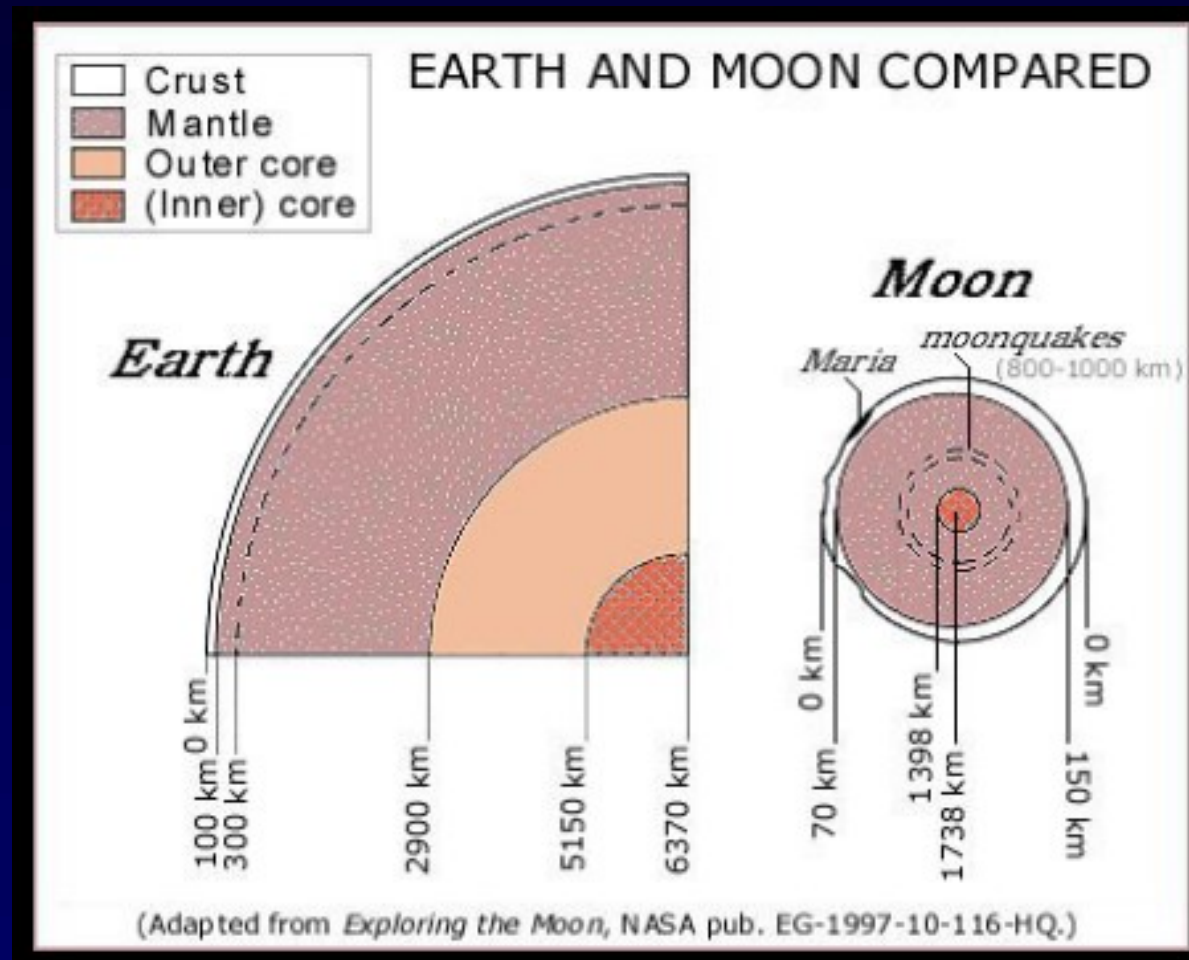
- **Evidences**

- 1. Moon rock is younger than other celestial bodies
- 2. Similar to composition to the earth
- 3. Moon is less dense than earth and other planets.



Computer simulated model

Internal Structure of the Earth and Moon



Like Earth, the Moon is a differentiated body. However, the exact nature of the internal structure of the Moon is still debated. The exact nature of the core (specific composition, size, liquid or solid) is still debated.

Origin of Earth's atmosphere

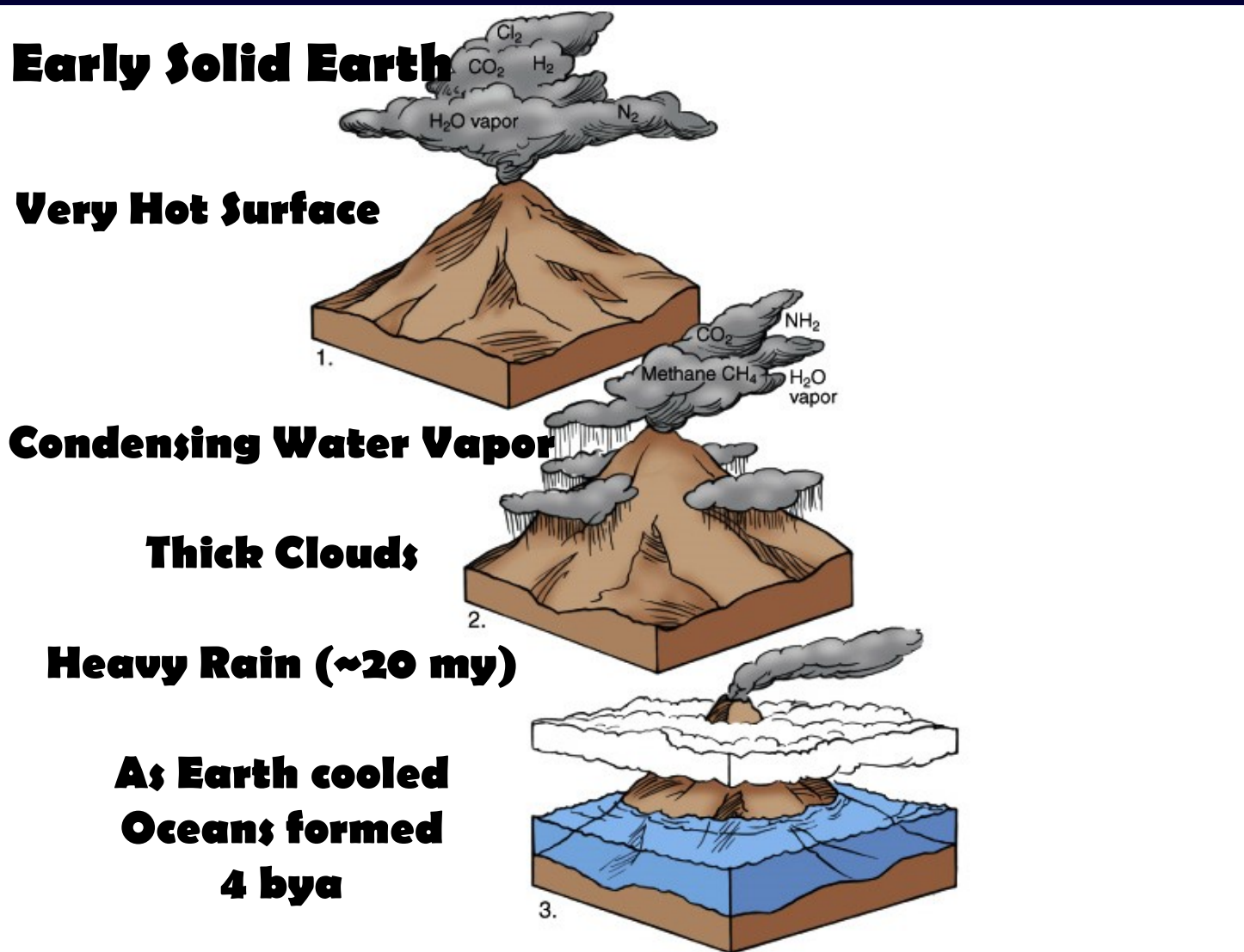
- Partial melting resulted in outgassing about 4 billion years ago
- Very different than today's atmosphere
- Similar to gases emitted from volcanoes
- Mainly water vapor (steam)
- Carbon dioxide, hydrogen, methane and ammonia



How are oceans formed?

- Most of the water on the planet probably erupted from huge volcanoes early in Earth's history.
- Volcanoes still produce a lot of water vapor, as well as other gases.
- A similar mixture would have formed Earth's first atmosphere.
- The water vapor turned into clouds that spilled torrential rain on the bare rocky surface of Earth's crust, flooding it to form the first oceans in ocean basins about 4 billion years ago.

Formation of the Atmosphere and the Ocean



- *Ice Comets were also important to adding water to the Earth system.*

Main constituents of Evolving Atmosphere



These would include:

Water vapor (H_2O)

Sulfur dioxide (SO_2)

Hydrogen sulfide (H_2S)

Carbon dioxide (CO_2)

Carbon Monoxide (CO)

Ammonia (NH_3)

Methane (CH_4)

Note that oxygen (O_2) gas is not created by volcanic eruptions

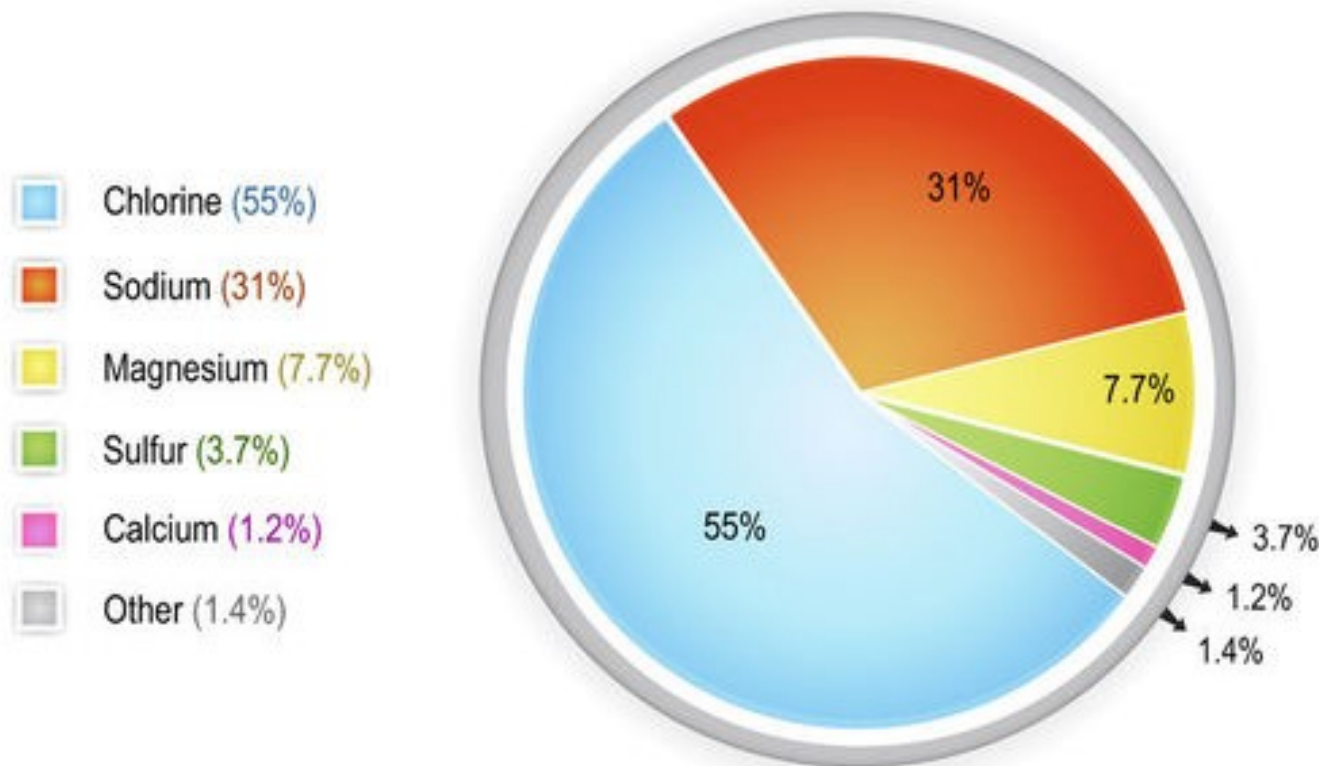
The origin of atmospheric oxygen on Earth

- Oxygen (O_2) production by **photosynthesis** is by far the dominant global process that replenishes atmospheric and oceanic oxygen essential to sustain all aerobic life.
- The O_2 on Earth was relatively scarce for much of our planet's 4.6 billion-year existence. But at some point, Earth underwent "the Great Oxidation Event (GOE)" for short, as ocean microbes evolved to produce O_2 via photosynthesis. O_2 first accumulated in Earth's atmosphere at this time and has been present ever since. It's been thought that this happened sometime between 2.5 and 2.3 billion years ago.
- The evolution of O_2 -producing cyanobacteria that use water as terminal reductant transformed Earth's atmosphere to one suitable for the evolution of aerobic metabolism and complex life.

The origin of atmospheric oxygen on Earth...Contd.

- The Great Oxidation Event was a time interval during which when the Earth's atmosphere and the shallow ocean first experienced a rise in the amount of oxygen. This began approximately 2.5 - 2.3 Ga (billion years) ago.
- Geological, isotopic and chemical evidence suggests that biologically-produced molecular oxygen (dioxygen, O₂) started to accumulate in Earth's atmosphere and changed it from a weakly reducing atmosphere practically free of oxygen into an oxidizing atmosphere containing abundant oxygen.

Main elements in Ocean Water



Ocean water is composed of many substances, many of them salts such as sodium, magnesium, and calcium chloride. The salt content is about 3.5% of the mass of ocean water, but the salt content or **salinity** is different in different locations in the Ocean.



Thank You

Pillow lava at CIR observed during manned submersible dive